

# Astra Arvis Navigation System

Orthopedic Imaging & Computer-Assisted Navigation · \$100K–\$500K  
Range

## EXECUTIVE LEAVE-BEHIND · FINANCIAL & CLINICAL SUPPLEMENT

This document is prepared as a supplemental resource to accompany the formal Value Analysis Committee capital equipment request for the Astra Arvis Orthopedic Navigation System. It provides peer-reviewed clinical evidence, facility-specific financial modeling frameworks, 2026 reimbursement context, and the regulatory background necessary for an informed committee review.

|                                                                        |                                                                   |                                                                     |                                                                       |                                                                    |
|------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>\$126K</b><br>REAL AFTER-TAX COST<br>\$200K SYSTEM<br>(37% BRACKET) | <b>\$4,721</b><br>HOPD VS. ASC<br>SPREAD<br>PER TKA (CMS<br>2026) | <b>9.0%</b><br>MALALIGNMENT<br>RATE W/<br>NAVIGATION<br>(VS. 31.8%) | <b>~7 mo</b><br>EQUIPMENT<br>PAYBACK<br>50<br>INCREMENTAL<br>CASES/YP | <b>20.4%</b><br>TKAS USING<br>ROBOTIC<br>ASSISTANCE<br>(AJRR 2023) |
|------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------|

### DOCUMENT TYPE

VAC Capital Equipment Supplement

### PREPARED

Q2 2026

### PRODUCT CATEGORY

Orthopedic Navigation & Imaging System

### APPLICABLE FACILITY TYPES

For-Profit Hospital · Non-Profit Hospital · Surgeon-Owned ASC

ASTRA · ARVIS

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## Executive Summary & VAC Recommendation

### RECOMMENDED ACTION FOR VAC APPROVAL

Approve capital purchase of the Astra Arvis Orthopedic Navigation System. The 2026 investment environment is uniquely favorable: permanent 100% bonus depreciation under the One Big Beautiful Bill Act reduces real net cost to 63 cents on the dollar for for-profit and surgeon-owned facilities, while the CMS CY 2026 Final Rule removes 285 musculoskeletal procedures from the Inpatient-Only list and adds 560 new ASC-billable procedures — creating a volume retention and revenue capture imperative that extends to non-profit systems regardless of tax position.

**\$126K**

NET AFTER-TAX  
COST  
\$200K SYSTEM,  
37%

**\$63K**

YEAR 1 TAX  
SAVINGS  
\$300K AT 21%  
CORP.

**\$4,721**

HOPD VS. ASC  
PER-TKA  
SPREAD

**~7 mo**

PAYBACK  
PERIOD  
50 CASES/YEAR

**9.0%**

MALALIGNMENT  
RATE  
NAVIGATION (VS.  
31.8%)

**Jun 30**

179D  
CONSTRUCTION  
START DEADLINE

#### FORCE 1: FEDERAL TAX LAW (OBBBA)

- › **100% bonus depreciation permanent** — One Big Beautiful Bill Act, PL 119-21, signed July 4, 2025
- › No phase-down, no sunset, no dollar cap
- › 2026 would have been a 20% bonus year under old law — OBBBA restored **\$50,400 in Year 1 value** on a \$300K purchase
- › Pass-through K-1 to surgeon-owners' personal returns in year of purchase
- › Finance as a loan (not lease) to preserve ownership and depreciation eligibility

#### FORCE 2: CMS CY 2026 COVERAGE EXPANSION

- › **285 MSK procedures removed from Inpatient-Only list**
- › **560 new ASC-eligible procedures** — largest single-year CPL expansion in history
- › New Level 7 Musculoskeletal APC for revision joints and complex fusions
- › Revision THA and TKA: newly billable in ambulatory settings for the first time
- › Navigation is a clinical prerequisite for performing these cases in an ASC setting

**VAC Decision Framework:** This is not a "nice to have" technology purchase. It is a capital investment with a documented 7-month payback, permanent tax advantage, and access to a newly-created procedure revenue category. Facilities that defer this decision defer the revenue; the tax benefit is permanent and does not expire.

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## Financial Analysis by Facility Type

ROI &amp; TAX IMPACT

### FOR-PROFIT HOSPITALS — CORPORATE TAX POSITION

| SCENARIO                                           | YEAR 1 DEDUCTION | YEAR 1 TAX SAVINGS | VS. OLD LAW      |
|----------------------------------------------------|------------------|--------------------|------------------|
| Old law (20% bonus — 2026 without OBBBA)           | \$60,000         | \$12,600           | —                |
| <b>OBBBA — 100% Bonus Depreciation (permanent)</b> | <b>\$300,000</b> | <b>\$63,000</b>    | <b>+\$50,400</b> |

**Bond covenant note:** GAAP and IRS depreciation run on separate books. Accelerating the tax deduction does *not* reduce GAAP EBITDA reported to bond rating agencies. No covenant exposure.

### SURGEON-OWNED ASC — PASS-THROUGH TAX SENSITIVITY (\$200K SYSTEM)

| OWNERSHIP % | K-1 DEDUCTION   | TAX SAVINGS (37%) | NET AFTER-TAX COST | EFFECTIVE COST |
|-------------|-----------------|-------------------|--------------------|----------------|
| 10%         | \$20,000        | \$7,400           | \$12,600           | 63¢            |
| 20%         | \$40,000        | \$14,800          | \$25,200           | 63¢            |
| <b>25%</b>  | <b>\$50,000</b> | <b>\$18,500</b>   | <b>\$31,500</b>    | <b>63¢</b>     |
| 50%         | \$100,000       | \$37,000          | \$63,000           | 63¢            |
| 100%        | \$200,000       | \$74,000          | \$126,000          | 63¢            |

At 37% bracket, every ownership stake pays 63¢ on the dollar. At 32%: 68¢. At 24%: 76¢. State tax savings are additive where state conforms to federal bonus depreciation. CA, NY, NJ do not conform.

### INCREMENTAL EBITDA & PAYBACK MODEL (SURGEON-OWNED ASC)

| SCENARIO         | ADDITIONAL CASES/ YEAR | BLENDED NET REVENUE | CONTRIBUTION MARGIN/CASE | INCREMENTAL EBITDA |
|------------------|------------------------|---------------------|--------------------------|--------------------|
| Conservative     | 25                     | \$8,500             | \$3,000                  | \$75,000           |
| <b>Base Case</b> | <b>50</b>              | <b>\$10,000</b>     | <b>\$4,500</b>           | <b>\$225,000</b>   |
| Optimistic       | 80                     | \$12,000            | \$6,500                  | \$520,000          |

Fixed costs excluded — incremental cases flow largely to contribution margin. Base case payback:  $\$126,000 \div \$225,000 = \sim 7$  months.

#### EQUITY VALUE CREATED (BASE CASE, 25% OWNER)

- › \$225K EBITDA × 9x multiple = **\$2.025M enterprise value added**
- › 25% owner's share: **\$506,250**
- › Net of \$31,500 after-tax cost: **+\$474,750**
- › Orthopedic ASC multiples: **7x–10x (2025–2026)**

### NON-PROFIT HOSPITAL — MIGRATION & VOLUME RETENTION MODEL (300 TKA PROGRAM)

| SCENARIO                                                 | HOPD<br>CASES | ANNUAL REVENUE (50/50<br>MIX) | ANNUAL REVENUE<br>IMPACT |
|----------------------------------------------------------|---------------|-------------------------------|--------------------------|
| All cases remain HOPD (navigation-<br>equipped)          | 300           | <b>\$6.30M</b>                | Baseline                 |
| 20% migrate to competitor ASC (60 cases)                 | 240           | \$5.60M                       | <b>(\$700K)</b>          |
| <b>35% migrate (1 high-volume surgeon<br/>recruited)</b> | <b>195</b>    | <b>\$4.97M</b>                | <b>(\$1.33M)</b>         |

**The surgeon recruitment math:** One high-volume joint surgeon performing 200 TKAs/year, migrating to a navigation-equipped ASC = **\$944,200/year in facility fee revenue permanently lost** (at \$4,721 HOPD premium × 200 cases). A \$250,000 navigation investment is recovered by retaining just 53 cases annually.

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## 2026 Reimbursement & CMS Coverage Landscape

The CY 2026 OPPS/ASC Final Rule represents the most significant expansion of ASC-eligible procedures in the program's history. The rule directly creates financial incentives for ambulatory navigation capability that did not exist in prior years.

### HOPD VS. ASC MEDICARE PAYMENT RATES (ESTIMATED CY 2026)

| PROCEDURE                            | CPT   | HOPD RATE (EST.) | ASC RATE (EST.) | HOPD PREMIUM   | NAVIGATION REQUIRED      |
|--------------------------------------|-------|------------------|-----------------|----------------|--------------------------|
| Total Knee Arthroplasty              | 27447 | ~\$14,005        | ~\$9,284        | ~\$4,721       | Yes — strongly preferred |
| Total Hip Arthroplasty               | 27130 | ~\$14,005        | ~\$9,478        | ~\$4,527       | Yes — strongly preferred |
| Revision TKA (NEW 2026 ASC-eligible) | 27487 | ~\$16,000+       | ~\$9,500–12,000 | ~\$4,000–6,000 | Yes — required           |
| Revision THA (NEW 2026 ASC-eligible) | 27134 | ~\$16,000+       | ~\$9,500–12,000 | ~\$4,000–6,000 | Yes — required           |
| Total Shoulder Arthroplasty          | 23472 | ~\$9,000         | ~\$8,700–9,200  | —              | Yes                      |
| ACL Reconstruction                   | 29888 | ~\$9,000         | ~\$4,500–5,500  | ~\$4,000       | Situational              |

Commercial payers run 150–300%+ of Medicare ASC rates for TJA in most markets. On-campus HOPD rates shown. Off-campus HOPDs acquired after November 2, 2015 subject to site-neutral payment (~40% of OPPS). Confirm campus designation before modeling revenue.

#### WHAT CHANGED IN CY 2026 (CMS FINAL RULE)

- › **285 MSK procedures removed from IPO list**
- › 560 surgical + 35 ancillary procedures added to ASC Covered Procedures List
- › New **Level 7 Musculoskeletal APC** for complex revisions and multilevel fusions
- › Revision THA and TKA: first-time ASC billing eligibility
- › Complex multilevel spinal fusions: selected CPTs (22551, 22554, 22612, 22630, 22633)
- › +2.6% ASC payment rate update

#### CRITICAL REIMBURSEMENT CLARIFICATION

- › CMS does **NOT** pay a facility premium for navigation-assisted TJA
- › Extremity navigation CPTs **0054T/0055T are bundled** under comprehensive APCs — not separately billable for TKA/THA
- › **Exception: Spine navigation CPTs 61781–61783 ARE separately billable** under HOPD OPPS — generate incremental billing on top of the spine fusion facility fee
- › The financial case for joint navigation is *volume, quality, and competitive position* — not a billing premium

**Market direction:** Sg2 Health Consultants projects **51% of total joints will be performed outpatient by 2026**. ASC joint replacement centers grew from 118 to 312 (2019–2023), with outpatient knee replacement volume up **293%** in four years. The site-of-service shift is not a prediction — it is already the dominant trend in orthopedic surgery delivery.

## Clinical Evidence Summary

PEER-REVIEWED DATA

The clinical argument for navigation is a financial argument. In bundled payment and DRG environments, revision avoidance translates directly to operating margin improvement without a corresponding second reimbursement. Peer-reviewed data consistently demonstrates a 46–83% relative reduction in key quality metrics across procedure categories.

### TOTAL KNEE ARTHROPLASTY: ALIGNMENT ACCURACY (META-ANALYSIS)

| METRIC                                     | NAVIGATION / CAS | CONVENTIONAL | RELATIVE REDUCTION | CLINICAL SIGNIFICANCE                   |
|--------------------------------------------|------------------|--------------|--------------------|-----------------------------------------|
| <b>Mechanical axis malalignment &gt;3°</b> | <b>9.0%</b>      | <b>31.8%</b> | <b>71%</b>         | <b>Primary driver of early revision</b> |
| Limb mechanical axis outliers              | 4.3%             | 18.6%        | 77%                | Component longevity                     |
| Femoral component outliers                 | 3.1%             | 18.4%        | 83%                | Sagittal plane alignment                |
| Tibial component outliers                  | 3.5%             | 12.2%        | 71%                | Coronal plane alignment                 |

### UKA REVISION RATE (AOANJRR 5-YEAR REGISTRY DATA)

|                             | MAKO ROBOTIC-ASSISTED      | CONVENTIONAL |
|-----------------------------|----------------------------|--------------|
| <b>5-Year Revision Rate</b> | <b>2.5%</b>                | <b>4.6%</b>  |
| 3-Year Revision Rate        | 2.5%                       | 5.5–5.8%     |
| Relative Reduction          | <b>46% fewer revisions</b> |              |

### LEARNING CURVE: CASES TO TIME NEUTRALITY

| SYSTEM TYPE             | CASES TO NEUTRALITY | ADD'L TIME (EARLY) |
|-------------------------|---------------------|--------------------|
| Mini robotic systems    | <b>5–7 cases</b>    | +6 min             |
| <b>Mako robotic TKA</b> | <b>9–15 cases</b>   | <b>Variable</b>    |
| CAS (optical tracking)  | 16–20 cases         | +10–20 min         |

"Navigation slows us down" is valid for 1–2 months only. The literature does not support it as a durable argument against adoption. A surgeon committed to working through 15–20 cases reaches full time efficiency.

### SPINE: DIRECT COST SAVINGS MODEL

| METRIC                                 | NAVIGATION                             | CONVENTIONAL | SAVINGS PER 100 CASES |
|----------------------------------------|----------------------------------------|--------------|-----------------------|
| Pedicle screw revision rate            | ~0%                                    | 3–4.4%       | —                     |
| <b>Cost per pedicle screw revision</b> | <b>\$23,762 (direct hospital cost)</b> |              | <b>\$71,286</b>       |

**LOS & SAME-DAY DISCHARGE**

- › Mako robotic TKA vs. conventional: **26% reduction in hospital length of stay** (prospective study)
- › Lower pain scores at 24, 48, 72 hours post-op; reduced opioid requirements
- › Faster recovery supports same-day discharge in ASC setting
- › Unexpected overnight admission rate: key accreditation quality metric for ASC joint replacement programs

**PAYER CONTRACTING LEVERAGE**

- › **Centers of Excellence designation** — Blue Distinction, BUCA networks direct volume to COE facilities; navigation outcomes data is a qualifying criterion
- › **Bundled payment advantage** — every avoided revision = \$20–50K in avoided cost within the bundle
- › **Direct employer contracting** — self-insured employers seek navigation-capable ASCs for TJA episodes (\$5,717 mean facility fee savings vs. HOPD; AJMC/Turquoise Health)

## Market Position & Competitive Context

### TECHNOLOGY ADOPTION

| MARKET METRIC                               | DATA POINT             |
|---------------------------------------------|------------------------|
| <b>Robotic TKA market share (end 2023)</b>  | <b>20.4%</b>           |
| Mako alone                                  | 10.3% of all TKA       |
| Growth rate 2017–2023                       | <b>+600% (sixfold)</b> |
| Navigation market CAGR (2025–2030)          | <b>13.9%</b>           |
| Global navigation market (2024)             | \$2.85 billion         |
| U.S. ASC orthopedic market (2030 proj.)     | \$15 billion           |
| % TJA outpatient projected 2026             | <b>51% (Sg2)</b>       |
| ASC joint replacement centers (2019 → 2023) | <b>118 → 312</b>       |

### COMPETITIVE POSITION REALITY

At 20.4% of TKA volume using robotic assistance — with sixfold growth in six years — this is no longer early-adopter technology. Facilities without navigation are increasingly asked by surgeons and patients why they don't have it.

69% of patients surveyed believe robotic-assisted surgery produces fewer complications and better outcomes (2021 patient survey). Consumer awareness has crossed the threshold where navigation-capable facilities have a measurable advantage in patient-choice markets.

### SURGEON RECRUITMENT: THE TRAINING COHORT EFFECT

Orthopedic surgery graduates completing residency and fellowship from 2018 onward trained on robotic and navigation platforms as a standard part of their education. These surgeons actively seek facilities with compatible technology as a condition of employment or privileges, and in documented cases have stated: "I will never do another joint replacement without robotics technology." This creates a structural recruitment advantage for navigation-equipped facilities that is not replicable by compensation alone.

The U.S. faces a projected shortage of **5,080 orthopedic surgeons by 2025** (AAOS projections). In a supply-constrained market, facilities offering navigation recruit from a larger eligible pool. Facilities without it recruit from what remains.

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## First-Time ASC Developer Reference

DE NOVO FUNDAMENTALS

| CATEGORY                                    | LOW<br>END    | HIGH<br>END     |
|---------------------------------------------|---------------|-----------------|
| Land / real estate                          | \$600K        | \$1.5M          |
| Construction / build-out                    | \$1.5M        | \$6.0M          |
| Medical equipment (2 ORs)                   | \$750K        | \$2.0M          |
| Pre-opening (legal, CON, consulting)        | \$200K        | \$500K          |
| <b>Working capital reserve <sup>△</sup></b> | <b>\$800K</b> | <b>\$1.5M</b>   |
| <b>Total project</b>                        | <b>\$4.0M</b> | <b>\$12.0M+</b> |

Working capital: budget 6–8 months operating costs as *liquid reserves*, not a credit line. Cash gap from opening to consistent collections is 4–6 months minimum.

### EQUIPMENT AT OPENING: DIRECT OPINION

- › \$300K Arvis in Year 1 → \$300K entity deduction
- › 4-surgeon group at 37%: **\$111K total tax savings** in your highest-cost year
- › 1-year delay NPV cost: ~\$8,880 across the group
- › **Finance with a loan, never a lease** — leasing forfeits the depreciation
- › Break-even: ~67 cases/month; two-OR standard = 1,500–1,700 cases/year

### FIVE FATAL MISTAKES

- › Underestimating working capital — 6–8 months, liquid
- › Starting payer contracting too late (begin 18 months before opening)
- › Phasing equipment post-opening — defers tax AND revenue
- › Passive physician investors
- › Oversizing for aspirational volume

## VAC Committee Reference: Objections & Responses

| COMMON OBJECTION                                              | EVIDENCE-BASED RESPONSE                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "Not in this year's capital budget"                           | Most \$100–500K revenue-generating purchases fall within CFO or VP-level authorization — no board cycle required. Frame as revenue retention: at \$5,000 blended differential per case, a \$250K system is covered by retaining 50 cases. Request mid-year approval with a one-page retention analysis using your facility's actual TKA volume.                                                                                           |
| "We need better ROI data before committing"                   | One variable dominates the ROI: committed cases per month from your surgeon group. 4 additional TJAs/month = \$480K/year in revenue. At \$126K after-tax cost, payback is under 4 months. Request a 30-minute utilization modeling session with your specific case volume and payer mix data.                                                                                                                                             |
| "Navigation slows down OR throughput"                         | The learning curve is real but short: 9–20 cases to time neutrality depending on the system. After the learning curve, OR times are equivalent to or faster than conventional. "It slows us down" is valid for the first 1–2 months only and is not a durable argument against adoption.                                                                                                                                                  |
| "We're evaluating a competitor system"                        | A comparison worth doing thoroughly. Evaluate total cost of ownership (service contract terms, upgrade cycle obligations), specific procedure categories each system enables vs. your case mix, and OR setup time at steady state. Surgeon-to-surgeon reference calls with current users of each system provide the most actionable comparison.                                                                                           |
| "There's no reimbursement premium for navigation"             | Correct — and any vendor claiming otherwise is overstating the case. CMS bundles extremity navigation CPTs (0054T/0055T) under comprehensive APCs. The financial case is: (1) navigation enables access to the highest-facility-fee procedure categories; (2) quality outcomes data drives payer contracting leverage; (3) competitive position at 20.4% robotic TKA market share. Spine navigation (61781–61783) IS separately billable. |
| "Let's wait and see how the reimbursement landscape develops" | The 2026 CMS rule has already restructured the landscape. 285 procedures came off the IPO list. 560 new ASC-eligible procedures are effective now. Waiting does not change these structural shifts — it only delays the facility's ability to perform and bill for the new procedure categories that require navigation.                                                                                                                  |

### VAC SUBMISSION CHECKLIST

**CLINICAL JUSTIFICATION (COMPLETE BEFORE SUBMISSION)**

- › Surgeon commitment letters / case volume attestations
- › Current malalignment rate data (if available) vs. 9.0% navigation benchmark
- › Revision rate data for TKA/UKA programs
- › LOS data for current joint replacement program
- › Peer-reviewed citations supporting navigation in your target procedure categories

**FINANCIAL DOCUMENTATION (COMPLETE BEFORE SUBMISSION)**

- › Custom utilization model with facility case volume + payer mix
- › GPO contract query (Vizient/Premier) — 10–27% off list documented
- › Financing method confirmation (loan vs. lease for depreciation eligibility)
- › CPA/tax advisor review of bonus depreciation applicability
- › Total cost of ownership: equipment + service contract + upgrade obligations

## Financial & Regulatory

### Disclaimer

**Important Notice:** This document is prepared as a consultative resource for informational and educational purposes only. It does not constitute — and should not be relied upon as — financial, tax, legal, accounting, regulatory, or investment advice of any kind.

#### NOT FINANCIAL OR TAX ADVICE

All financial analyses, tax scenarios, depreciation calculations, and ROI projections are illustrative estimates based on publicly available data, published tax rates, and industry benchmarking. Tax laws, including provisions of the One Big Beautiful Bill Act (PL 119-21, July 2025) and all IRS guidance, are subject to change. Individual outcomes depend on entity structure, ownership percentages, income levels, passive activity rules, state conformity, and specific transaction terms.

**All recipients must consult their own qualified CPA, tax attorney, or financial advisor before making any capital investment or financing decision.**

#### REIMBURSEMENT VARIABILITY

Reimbursement figures are based on CMS published payment rules, MedPAC reports, and industry benchmarking as of Q2 2026. Actual reimbursement varies by local coverage determinations, hospital wage indices, DSH adjustments, individual payer contracts, procedure coding, and annual CMS rule updates. Nothing herein guarantees any specific reimbursement outcome for any facility or procedure.

#### REGULATORY COMPLIANCE

This document does not address compliance with the Stark Law (42 U.S.C. §1395nn), Anti-Kickback Statute (42 U.S.C. §1320a-7b), False Claims Act, or state healthcare regulatory requirements. Physician ownership of ASCs and related financial relationships are subject to complex regulatory requirements. Engage qualified healthcare legal counsel before any ownership, investment, or equipment financing arrangement.

#### STATE LAW VARIANCE

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Package prepared Q2 2026 · Tax and reimbursement data current as of date of preparation  
Verify all figures with primary sources and qualified advisors prior to use in any financial decision

**ASTRA · ARVIS**